

PRAVEEN N

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EXECUTIVE SUMMARY

Computer Science student (AIML specialization) at SRM University with hands-on experience in software development, data-driven systems, and building end-to-end technical solutions. Demonstrated ability to work across the full project lifecycle — from processing raw satellite data to deploying models achieving 87% accuracy. Complementary skills in full-stack web development including real-time applications. Passionate about using technology and data to solve real-world problems at scale.

SKILLS

Programming Languages: Python, C, JavaScript, TypeScript

Web Technologies: React, Node.js, Express.js, Tailwind CSS, WebRTC, Socket.IO, Flask (basic), HTML, CSS

Databases: MySQL, PostgreSQL

Cloud & DevOps: AWS (EC2, S3), Docker, Kubernetes, VPC, Subnets

Data Science & Analysis: Pandas, NumPy, Matplotlib, Tableau, Excel

Version Control & Tools: Git, GitHub

Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, U-Net CNN, ConvLSTM, MediaPipe, Model Evaluation

Computer Vision: OpenCV, Facial Landmark Detection, Satellite Image Processing

EDUCATION

B.Tech Computer Science and Engineering w/s AIML — CGPA: 7.8/10

SRM Institute of Science and Technology, Kattankulathur 2023 – 2027

PROJECTS

Fishing Zone Recommendation System Using Deep Learning | *Python, Pandas, NumPy, PyTorch, U-Net CNN, ConvLSTM, Satellite Data Analytics*

- Processed 6 oceanographic parameters (SST, SSH, velocity, chlorophyll, plankton) from Copernicus satellite data (2018–2025) across 15,000+ images; achieved 87% accuracy and IoU of 0.78 — 20% improvement over traditional methods.
- Designed a ConvLSTM-based spatiotemporal model capturing temporal ocean dynamics; built U-Net CNN segmentation pipeline for precise fishing zone boundary detection.
- Created interactive visualization dashboard with color-coded fishing zone maps and GPS coordinates, demonstrating ₹7–10 lakhs potential annual savings per vessel through AI-driven zone targeting.

Driver Drowsiness Detection | *Python, MediaPipe, OpenCV, Computer Vision*

- Built a real-time computer vision pipeline using MediaPipe facial landmark detection and eye-blink ratio (EAR) analysis to classify driver fatigue with high reliability.
- Triggered real-time audio/visual alerts on drowsiness detection; improved model robustness using MediaPipe's multi-face feature extraction across varied lighting conditions.

SRM Connect — Real-Time Chat & Video Platform | *React, Node.js, Supabase, WebRTC, Socket.IO*

<https://srmconnect2025.vercel.app>

- Built a full-stack real-time communication platform for SRM students with Supabase-backed authentication, live chat, and random WebRTC video-chat matching.
- Optimized peer-to-peer WebRTC connections for low-latency video calls; designed a responsive UI for seamless cross-device experience.

CERTIFICATIONS

- EY – EY Technology Risk Job Simulation
- Deloitte Australia – Data Analytics Job Simulation
- Tableau and Data Analysis
- Introduction to Business Intelligence (Infosys Springboard)
- Go for Gold Challenge – Gold Level, Accenture